

Name: _____

Date: _____

PYTHAGORAS, (FINDING A SHORTER SIDE)

LO: I can use Pythagoras' theorem to find a shorter side of a right-angled triangle.

Finding a shorter side

When the **hypotenuse** and one shorter side are known, rearrange Pythagoras to find the other: $a^2 = c^2 - b^2$.

Remember: Always subtract the known shorter side squared from the hypotenuse squared. The hypotenuse is the longest side.

Quick examples

●	$c = 13, b = 5: a = (169 - 25) = 144 = 12$	subtract then root
●	$c = 10, b = 6: a = (100 - 36) = 64 = 8$	rearranged formula
●	$c = 15, b = 9: a = (225 - 81) = 144 = 12$	Pythagorean triple
●	$c = 13, b = 5: a = (169 + 25)$	must subtract, not add

Key idea

Rearrange $a^2 + b^2 = c^2$ to $a^2 = c^2 - b^2$, then take the square root.

$$a = \left(c^2 - b^2 \right)$$

Subtract known shorter side squared from hypotenuse squared

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - integer answer

Find side a if $c = 10$ and $b = 8$.

$$a^2 = c^2 - b^2 = 100 - 64 = 36$$

$$a = 36$$

$$a = 6$$

$$a = 6$$

Always square both known sides first, then subtract.

Example 2 - non-integer answer

Find side a if $c = 9$ and $b = 5$.

$$a^2 = 81 - 25 = 56$$

$$a = 56$$

$$a = 7.48 \text{ (2 d.p.)}$$

$$a = 7.48$$

If the answer is not a perfect square, round appropriately.

Example 3 - in context

A rope 17 m long is tied to the top of a 15 m pole and pulled taut to the ground. How far from the base is it?

Hypotenuse (rope) = 17, vertical

(pole) = 15

$$d^2 = 17^2 - 15^2 = 289 - 225 = 64$$

$$d = 64 = 8 \text{ m}$$

$$8 \text{ m}$$

Identify the hypotenuse (always the longest side), then subtract.

Example 4 - checking which is the hypotenuse

A triangle has sides 20 cm, 16 cm, and an unknown third side. The 20 cm side is the hypotenuse. Find the third side.

$$a^2 = 20^2 - 16^2 = 400 - 256 = 144$$

$$a = 144 = 12$$

Third side = 12 cm

$$12 \text{ cm}$$

The hypotenuse is always stated or identified as the longest/opposite the right angle.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding a (integer answers)

Find the missing shorter side.

- a) $c = 5$, $b = 3$
- b) $c = 13$, $b = 12$
- c) $c = 15$, $b = 12$
- d) $c = 25$, $b = 7$

2. Finding a (decimal answers)

Find the missing side to 1 d.p.

- a) $c = 8$, $b = 5$
- b) $c = 12$, $b = 7$
- c) $c = 10$, $b = 4$
- d) $c = 20$, $b = 11$

3. In context

Solve each problem.

- a) Ladder 10 m, reaches 8 m up wall.
Distance from wall?
- b) Kite string 26 m, horizontal distance 10 m. Height?
- c) Ship sails: total journey 17 km, east 15 km. North?
- d) Slide 13 m, platform height 5 m.
Ground distance?

4. Mixed triangles

Find x in each right-angled triangle.

- a) Hypotenuse 17, one side x , other side 15
- b) Hypotenuse 26, one side x , other side 24
- c) Hypotenuse 20, one side x , other side 12
- d) Hypotenuse 29, one side x , other side 21

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To find a shorter side, rearrange to $a^2 = c^2 - b^2$ ☐

Correct answer: _____

Reason: _____

b) $a = c - b$ gives the missing side ☐

Correct answer: _____

Reason: _____

c) The hypotenuse is always the longest side in a right-angled triangle ☐

Correct answer: _____

Reason: _____

d) $a^2 = c^2 + b^2$ when finding a shorter side ☐

Correct answer: _____

Reason: _____

e) If the answer is not exact, round to an appropriate degree of accuracy ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A right-angled triangle has hypotenuse $(2x + 1)$ cm and legs x cm and $(x + 7)$ cm. Use Pythagoras to form an equation and find x . Hence find all three side lengths.

Expression: _____

Simplified: _____

b) Two vertical poles are 8 m and 14 m tall, standing 8 m apart on level ground. A wire connects the tops of the two poles. Find the length of the wire.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

